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Pathway analogy for computational fluid dynamics simulations of fluid flow through structured packed beds

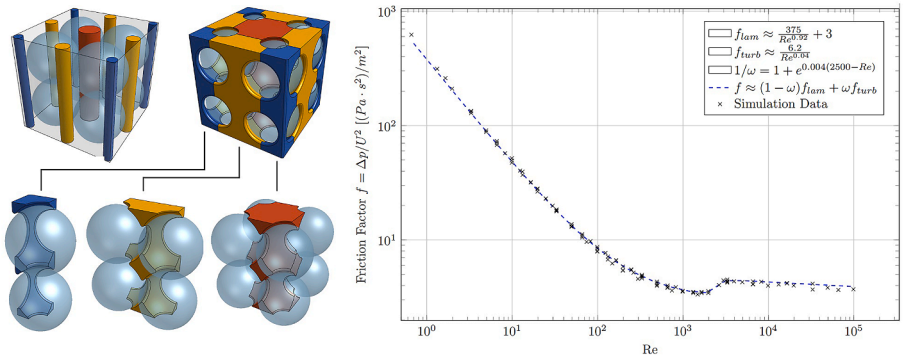
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HIGHLIGHTS

- Novel pathway analogy decomposes structured packed bed flow.
- Cross-flow between pathways is negligible (<1%).
- Universal friction factor correlations independent of particle size.
- Mechanistic model sums weighted pathway contributions.
- Validated against experimental data across wide Re range.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

Predicting pressure drop in structured packed beds is critical for the design of industrial chemical reactors and heat exchangers. While particle-resolved Computational Fluid Dynamics (CFD) offers high fidelity, its prohibitive computational cost for large-scale systems necessitates efficient modeling strategies. This study introduces a novel pathway analogy for fluid flow through structured packed beds in a simple cubic arrangement. The approach deconstructs the complex interstitial flow field into a finite set of discrete, hydraulically independent pathways—classified as full, half, or quarter based on their proximity to the container walls. High-fidelity CFD simulations confirm that cross-flow between these pathways is negligible (< 1%) both in laminar and turbulent regimes, justifying their treatment as independent flow units. Universal correlations for the friction factor and a normalized pressure drop are derived for each pathway type as a function of Reynolds number, demonstrating negligible dependence on particle diameter. These foundational correlations are synthesized into a mechanistic engineering model that reconstructs the total pressure drop for a bed of any dimension by summing the weighted contributions of all individual pathways. The model is rigorously validated against experimental data from the literature, showing strong agreement across a wide range of Reynolds numbers and bed configurations. This pathway analogy provides a robust, physics-based framework for drastic computational domain reduction, offering a valuable tool for the efficient design and scaling of structured packed bed systems.

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1. Introduction

The study of fluid flow through packed beds is a cornerstone of chemical and process engineering, underpinning the design and optimization of ubiquitous industrial units such as chemical reactors, heat exchangers, and filtration systems (Sadowski et al., 2024; von Seckendorff & Hinrichsen, 2021; Zamengo et al., 2023). In these applications, the packed bed enhances transfer processes by providing a large surface area for interfacial contact. A critical parameter governing the energy efficiency and operational cost of such systems is the pressure drop across the bed, which is intrinsically linked to the complex interstitial flow patterns around the packing elements (Ambekar et al., 2022; Zawadzki & Blatkiewicz, 2023; Zhang et al., 2022).

There are several packing methods actively used in industry, which are grouped into two large groups: random and structured. The first group is more suitable for cost-effective applications that operate with clean working fluids at the cost of higher pressure drop that is usually aggravated by packing element attrition over time. The second group is superior for highly controllable and scalable applications that require the lowest hydraulic resistance of the packing. The considered cubic packing is common for cross-flow configurations of reactors with rectangular contact vessels that are used for dehumidification (Lawal et al., 2018; Liu et al., 2006), thermal/chemical storage (Yu et al., 2023; Yu et al., 2025), and other applications.

Predicting this pressure drop accurately remains a significant challenge, particularly when bridging the gap from small-scale, well-characterized systems to industrial-scale apparatuses. Random packings are often treated as porous materials with corrections for wall effects, whereas structured beds are mostly considered as a set of flow channels (Georgousis et al., 2025; Nikitin et al., 2025).

Computational Fluid Dynamics (CFD) has emerged as a premier tool to unravel the intricacies of these flows, providing high-fidelity, particle-resolved insights into velocity distributions and pressure loss mechanisms (Dixon & Partopour, 2020; Lanade & Hassan, 2024; Zahir et al., 2023). The comprehensive review by Dixon and Partopour (Dixon & Partopour, 2020) underscored the dual utility of CFD: it can serve for direct simulation and for generating crucial closure parameters and physical insights for simpler, more tractable engineering models. This highlights a central theme in the field—the perpetual trade-off between the high accuracy of fully-resolved simulations and their prohibitive computational cost. Resolving the flow around each particle typically demands on the order of one million computational cells per 50 particles (Dixon et al., 2011; Zhang et al., 2022), rendering simulations of industrial-scale beds containing hundreds of thousands of particles intractable. This has driven the development of strategies for domain reduction and model simplification.

A widespread approach to managing computational expense is the use of volume-averaged methods, such as those employing the Ergun equation (Ergun, 1952) or other equations (Macdonald et al., 1979; Nikitin et al., 2025; Papkov et al., 2024), which describe the pressure drop as a superposition of viscous and inertial losses. While invaluable for initial design, these porous medium models inherently average local physics and often fail to accurately capture the significant flow heterogeneities, especially pronounced wall effects, and the specific morphology of the packing. To overcome these limitations, many researchers have turned to unit cell and representative elementary volume (REV) concepts. For instance, Calis et al. (Calis et al., 2001) conducted a combined CFD and experimental study on a novel structured catalytic reactor packing, successfully validating pressure drop and flow profiles. Their work demonstrated the value of CFD for specific structured designs but was tied to a particular geometry. Similarly, Bai et al. (Bai et al., 2009) coupled the Discrete Element Method (DEM) with CFD to simulate flow in randomly packed beds, capturing the essential randomness but still at a high computational cost that scales with system size.

For structured packings, which offer the advantage of predictable and reproducible geometry, the strategy of identifying a minimal representative unit becomes particularly powerful. Studies on structured beds, such as those by Petre et al. (Petre et al., 2003) on heat transfer and Wang et al. (Wang et al., 2018) on grille-sphere composite packings, have effectively utilized both experimental and numerical methods to characterize specific configurations. These works provide valuable data and correlations for their specific geometries.

However, they often treat the wall effect as an external correction factor or a boundary condition to be empirically fitted rather than deriving it from a fundamental decomposition of the flow field. While the classic Ergun equation is widely used to predict pressure drops in packed beds, the significant wall effects present in beds with low tube-to-particle diameter ratios necessitate modified correlations, though existing ones often disagree and fail for very low ratios (Cheng, 2011). For such bounded beds, it is established that the wall increases pressure drop at low Reynolds numbers but decreases it at high Reynolds numbers compared to unbounded beds, with recent particle-resolved CFD (PRCFD) simulations showing that common corrections overestimate the effect in the viscous regime (Dixon, 2024). Furthermore, comprehensive numerical studies encompassing various flow regimes and packing structures have validated simulation models by achieving strong agreement with experimental friction factor data, helping to define the applicable Reynolds number range for the Ergun regime and quantify the impact of bounding walls (Gorman et al., 2019).

A more physics-based approach was suggested by the concept of “flow pathways” or “interstitial networks,” where the global flow is considered a sum of flows through discrete channels (George et al., 2023; Nishida et al., 2025). While this concept is occasionally invoked qualitatively, a quantitative, generalized framework for deconstructing a packed bed into its constituent hydraulic pathways – each with a characterized pressure drop response – and then reconstructing the total pressure drop for any bed width has remained underdeveloped. Furthermore, the analysis of complex flows through porous structures and wavy channels is also pivotal in microfluidic applications, particularly in electroosmotic flow, where anisotropic porosity and varying surface conditions significantly influence the flow dynamics (Khan & Sasmal, 2023; Tripathi et al., 2020).

This work addresses the challenge of lateral domain reduction for structured packed beds arranged in a simple cubic configuration. We introduce a novel “pathway analogy” that conceptualizes the flow not as a continuum through a porous medium but as a superposition of discrete flows through a network of well-defined, repeating interstitial channels. These pathways – classified as full, half, or quarter based on their proximity to the container walls – form the fundamental building blocks of the flow field. We demonstrate that for turbulent flow regimes typical of industrial applications, cross-flow between these adjacent pathways is negligible (less than 1%), justifying their treatment as hydraulically independent.

The primary objective of this study is to develop a robust and efficient methodology for predicting the pressure drop in rectangular structured packed beds by characterizing the hydraulic behavior of these individual pathway units. Through a comprehensive set of CFD simulations, the generalized correlations for the friction factor and normalized pressure drop for each pathway type, as a function of the Reynolds number, were derived. These foundational correlations are then synthesized into a direct engineering correlation that estimates the total pressure drop for any given bed dimensions and particle size. The proposed model is validated against experimental data from the literature, showing strong agreement across a wide range of conditions.

2. Pathway concept

The most basic configuration of structured packings is simple cubic. It consists of an orthogonal lattice of equally sized spheres, where each sphere is in contact with its nearest neighbors at six contact points. Fluid flows through such structured packings via some complex-shaped

channels or pathways. The simplest way to visualize these pathways is to consider eight spheres arranged in an 2 × 2 × 2 orthogonal lattice. An interstitial space forms at the intersection of these spheres, creating a concave octahedral void. That is the core element of the central pathway that can be chained streamwise when describing a full-length column of the packed bed, as shown in Fig. 1.

While the flow is not strictly confined to these pathways and can migrate freely to adjacent channels, the streamwise pressure gradient exceeds the transverse gradient by four orders of magnitude in turbulent flows. The orifice flow equation that is thoroughly validated for submerged jets (White, 2016) can be used to estimate a cross flow rate between adjacent pathways:

$$G^m_{cross} = C_d \cdot A_{perf} \cdot \sqrt{2\rho_f |\Delta P_{loc}|}, \tag{1}$$

where C_d is the discharge coefficient (accounts for flow contraction, viscosity, and turbulence effects), A_{perf} is the cross-sectional area of perforations, ΔP_{loc} is the local pressure difference between the two adjacent channels at each perforation, and ρ is the fluid density. Here we preserve the term "perforation" for consistency with the orifice flow equation.

Hence, less than 1% of the flow diverges from its primary pathway: $\frac{G^m}{G^m_{cross}} \propto \sqrt{\frac{\Delta P}{\Delta P_{loc}}}$. This was confirmed during the preliminary simulations, as shown later in the Simulation Results section. Although crossflows may become more substantial at Re < 50, such conditions occur infrequently in industrial applications. Thus, the cross flow was considered negligible in the present study.

In a column with a finite number of particles in each layer (original layout composition), pathways near the external walls consist of halves and quarters of the described polyhedron, as shown in the bottom row of shapes in Fig. 2. The number of pathways in a rectangular column can be expressed in terms of the row count m and the column count n in each layer:

$$\begin{cases} N_1 = (m-1)(n-1); & \text{full pathways} \\ N_{1/2} = 2(m+n-2); & \text{half pathways} \\ N_{1/4} = 4; & \text{quarter pathways} \\ N_{\sum} = mn + m + n + 1. & \text{all pathways} \end{cases} \tag{2}$$

These hold true for any aspect ratio of the rectangular layout of spherical particles, from unity (square) to max(m, n).

Assuming a plug flow impinging on the packed bed, the initial pressure distribution at the front layer of particles is uniform, implying hydraulically balanced parallel flow through the pathways. This yields equality of the absolute values of pressure drop across all pathways, with the minor discrepancy mentioned above. Meanwhile, the velocity and flow rate across individual pathways vary. Hence the total pressure drop within the packing can be decomposed as follows:

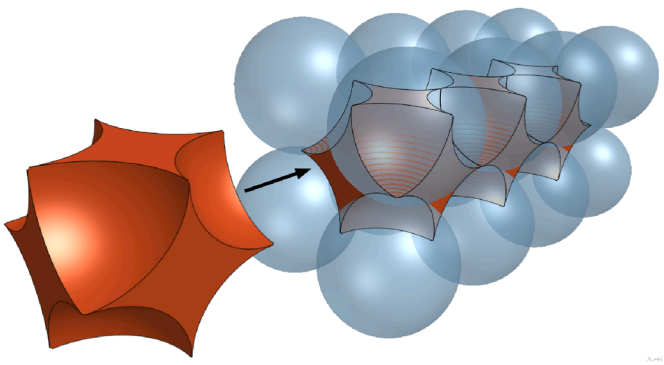


Fig. 1. Pathway concept representing the flow channel as a set of chained interstitial voids.

$$\Delta P = \frac{N_1 \Delta P_1 + N_{1/2} \Delta P_{1/2} + N_{1/4} \Delta P_{1/4}}{N_{\sum}} \approx \Delta P_i, \tag{3}$$

where $\Delta P_1 \approx \Delta P_{1/2} \approx \Delta P_{1/4}$ are the pressure drops in each pathway.

3. Numerical simulation

Computational fluid dynamics (CFD) simulations were performed in ANSYS Fluent to generate a large dataset, from which analytical correlations between pressure drop, particle size, and fluid flow characteristics were derived.

The following basic formulation of the Navier-Stokes equations was used:

$$\nabla \cdot (\rho u) = 0; \tag{4}$$

$$(u \cdot \nabla) u = -\frac{1}{\rho} \nabla p + \nu (\nabla^2 u) + \nabla R_{ij}; \tag{5}$$

$$R_{ij} = \nu_t (\nabla^2 u) - \frac{2}{3} k \delta_{ij}. \tag{6}$$

Mass flux divergence in the continuity Eq. (4) was set to zero due to mass conservation. The momentum Eq. (5) was used in a steady-state formulation. The system was considered isothermal; thus, the convective heat transfer was eliminated, and the gravity term was excluded from the equation set. The Reynolds stress (Eq. (6)) term ∇R_{ij} had a typical formulation adopted from the $k - kl - \omega$ turbulence model proposed by Walters and Cokljat (2008) in the late 2000s. This model was selected based on the success of previous implementations (Nikitin & Pashchenko, 2025).

At the inlet, a uniform velocity profile was set with the Dirichlet boundary condition: $u(x) = U_0 \vec{n}$ and turbulent intensity $I = \frac{5.5}{Re^{0.0407}}$. The Reynolds number was calculated regarding the particle diameter instead of the whole column to account for the porosity effects:

$$\mathrm{Re} = \frac{\rho_f U D_p}{\mu_f}. \tag{7}$$

At the outlet, a surface-normal zero gradient of pressure $(\partial p / \partial \vec{n})$ with $p_{\text{gauge}} = 0$ was set. Fluid-solid interfaces for column walls and particle walls were treated differently. The column walls were set with a null friction factor to eliminate the influence of the entrance and exit regions of the computational domain. The particles were considered hydraulically smooth; thus, their walls were set with a no-slip boundary condition.

The flow within the considered packed beds was subsonic, with a maximum Mach number below 0.3. However, an anticipated pressure drop of over 5% meant that the working medium — air at 25 °C — was considered moderately compressible. To eliminate the energy equation from the mathematical formulation, the ideal gas law was replaced with a linear correlation between static pressure and fluid density: $\rho(p) = 1.19 + 0.00235p$. The viscosity of the working fluid was set to $1.81 \cdot 10^{-5}$ Pa·s.

A routine verification procedure of the simulation results was performed during each simulation run. It included continuous control of the residuals and the pressure drop. The following stopping criteria were applied: $E_{\Delta P} = 0.001$, $E_{\text{continuity}} = 0.01$, $E_{\text{velocity}} = 10^{-5}$, and $E_{\text{pressure}} = 10^{-6}$.

Each pathway type (full, half, and quarter) was represented by a separate computational domain. Each domain was a set of 16 chained reentrant polyhedrons, as shown in Fig. 2. Entrance and exit regions of length $5D_p$ were created to eliminate the effects of open boundaries on the flow inside the packed bed, as shown in Fig. 3.

The following cases for the pathways were considered in the present study to bound superficial velocity within the 0.01–100 m/s range while keeping a wide Reynolds range of 0.1–10^5:

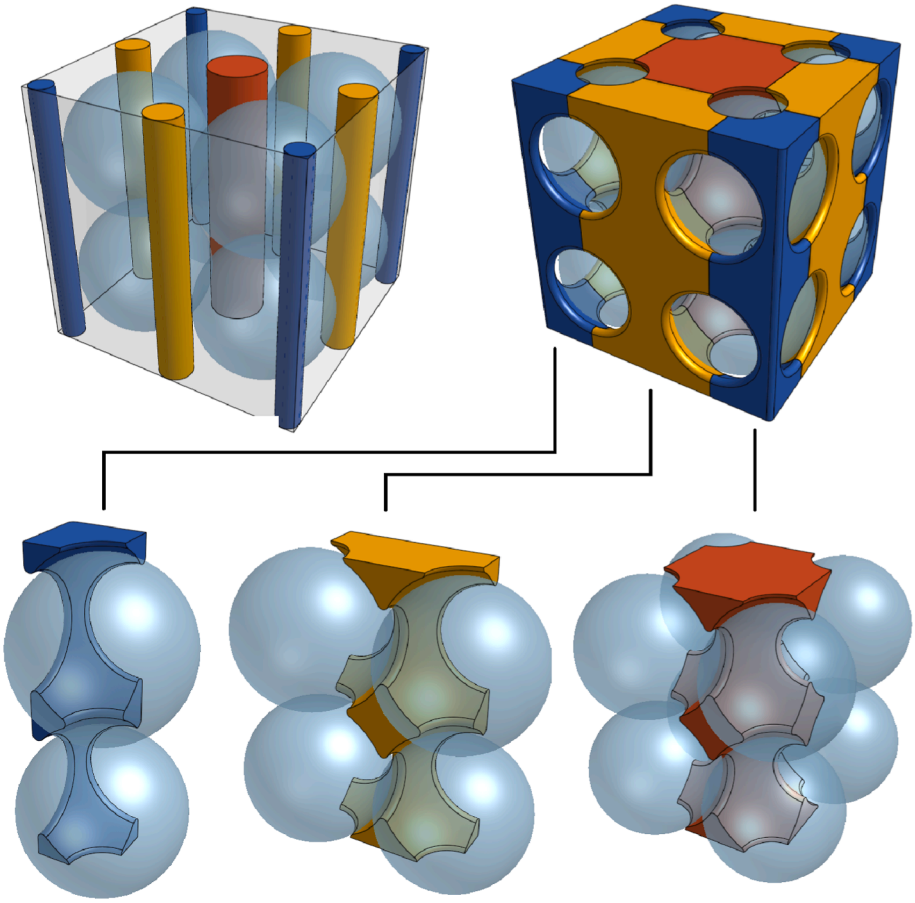


Fig. 2. Pathway profiles: circular (top left) and reentrant polyhedron (top right).

- Full pathways ($PW-1$) with particle diameters $D_p = \{1, 2.5, 5, 10, 15\}$ mm;
- Half pathways ($PW-1/2$) with particle diameters $D_p = \{1, 5, 15\}$ mm;
- Quarter pathways ($PW-1/4$) with particle diameters $D_p = \{1, 5, 15\}$ mm.

The computational domains were discretized using unstructured meshes comprised of nonuniform polyhedral cells with a growth rate of 1.2. Linear cell size was limited to $\Delta x_{\max} = 0.5$ mm. Up to five prismatic layers with a growth rate of 1.5 were applied to fluid-solid interfaces on spherical voids to enhance solution versatility in high-gradient regions. First layer thickness, growth rate, and number of layers varied across the mesh to ensure wall distance $y^+ \sim 1$ and minimize the stepping of prismatic layers.

The quality of the meshes was assessed using orthogonality and skewness factors that appeared to be within viable ranges, as shown in Table 1. Here, the particle diameter D_p is specified in parentheses. Cell count N_{cell} is given in millions of cells. Cell orthogonality was calculated as the cosine of the maximum angle between each cell face's normal vector and the vector connecting the cell's centroid to the adjacent cell's centroid. Cell skewness was calculated as the volume difference between that cell and its isometric (equilateral) equivalent.

One can clearly see that larger particles provide noticeably better meshes, both in terms of orthogonality and skewness. This can be explained by a higher ratio of volume to surface area. The average cell size approaches asymptotically the maximum cap $\Delta x_{\max} = 0.5$ mm as the meshing algorithm performs spatial discretization based on surface curvature and node proximity.

The mesh dependence of the simulation results was estimated using Richardson's extrapolation technique, the graphical representation of which is shown in Fig. 4. $PW-1$ ($D_p = 5$ mm) case with flow superficial

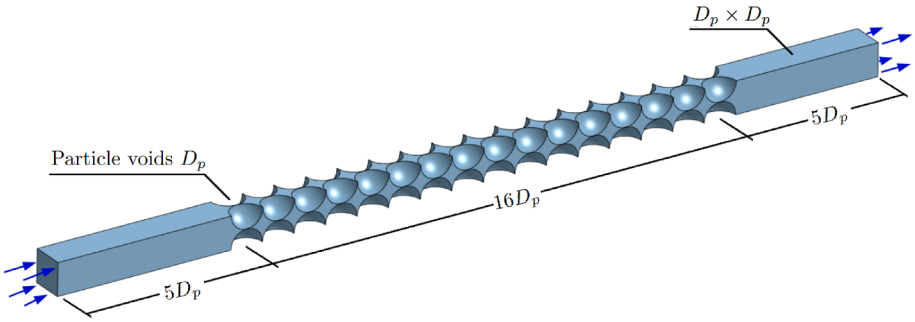


Fig. 3. Computational domain for a full pathway.

Table 1
Mesh quality metrics.

Case	N_{cell}	Δx [mm]	Avg Ortho	Min Ortho	Avg Skew	Max Skew
$PW-1$ ($D_p = 1$ mm)	0.22	0.04	0.85	0.15	0.15	0.85
$PW-1$ ($D_p = 2.5$ mm)	0.26	0.1	0.82	0.16	0.17	0.84
$PW-1$ ($D_p = 5$ mm)	0.27	0.2	0.82	0.2	0.17	0.8
$PW-1$ ($D_p = 10$ mm)	0.49	0.3	0.9	0.2	0.1	0.97
$PW-1$ ($D_p = 15$ mm)	1.1	0.4	0.93	0.19	0.07	0.81
$PW-1/2$ ($D_p = 1$ mm)	0.17	0.04	0.81	0.15	0.17	0.85
$PW-1/2$ ($D_p = 5$ mm)	0.2	0.2	0.79	0.19	0.2	0.81
$PW-1/2$ ($D_p = 15$ mm)	0.67	0.4	0.89	0.2	0.09	0.8
$PW-1/4$ ($D_p = 1$ mm)	0.11	0.03	0.79	0.15	0.19	0.85
$PW-1/4$ ($D_p = 5$ mm)	0.13	0.2	0.76	0.2	0.22	0.8
$PW-1/4$ ($D_p = 15$ mm)	0.41	0.3	0.85	0.2	0.11	0.8

velocity (100 m/s) was simulated on six meshes with an average cell size Δx ranging from 0.11 to 0.74 mm. The simulation results were extrapolated to a theoretical mesh that has $\Delta x \to 0$. The pressure drop on that mesh was estimated to be 664 kPa. Extrapolation indicated that meshes with an average cell size of up to 0.25 mm were within the accepted tolerance range ±1%.

4. Simulation results

4.1. Cross-flow between pathways

Preliminary simulations were conducted to validate the analytical assumption regarding the negligibility of cross-flow between pathways, an assumption supported by Eq. (1). The simulations were performed for a packing of 10 mm diameter particles arranged in a $2 \times 2 \times 16$ array using the CFD model described previously. Two flow regimes were tested: laminar (Re = 1000, $U_{in} = 1.5$ m/s) and turbulent (Re = 10000, $U_{in} = 15$ m/s). Velocity fields, shown in Fig. 5, indicate substantial spikes in the transverse velocity near the contact points between adjacent particles, but not within the interstitial voids. These spikes may result from pressure singularities that arise at these contact points, as the velocity magnitudes are directly proportional to the superficial velocity and decay monotonically in the streamwise direction along with the pressure gradient. When the singularity-induced artifacts are excluded, the transverse velocity component was found to be negligible in both the laminar and turbulent flow configurations.

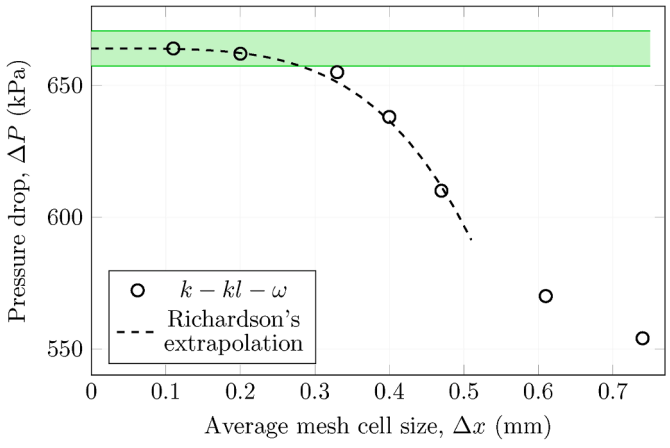


Fig. 4. Mesh dependency analysis for the case $PW-1$ ($D_p = 5$ mm) indicating maximum cell size $\Delta x_{max} = 0.26$ mm that satisfies Richardson's condition of grid independence.

4.2. Data verification

Airflow of variable velocity through three types of pathways within beds packed with spherical particles of various sizes yielded $21 \times (5 + 2 \times 3) = 231$ cases of parametric study. These cases were grouped into series with varying velocities. Several techniques were used to evaluate data smoothness, continuity, and coherence.

The simulated data series were compared with the corresponding Ergun correlations (Ergun, 1952) and the available experimental data. The overall discrepancies were below 5%, with an average deviation of 2.7%. The correlation between pressure drop ΔP and superficial flow velocity U followed a polynomial form consistent with the Forchheimer equation:

$$\Delta P = a \cdot U + b \cdot U^2, \tag{8}$$

where the coefficients a and b were individually fitted for each data series. Residuals of each simulated data series were randomly scattered around $Res = 0$ without any systematic patterns. This indicated the smoothness of the simulated data and an absence of discontinuities.

Simulation data represented in the form of friction factor $f = \Delta P/U^2$ clearly follow the trends from the Nikuradse experiments on fluid flow in rough pipes (Nikuradse, 1933) as shown in Fig. 6. The friction factor for the pathways has two distinct correlations for laminar and turbulent regimes, with a characteristic transition region at $2000 \leq \mathrm{Re} \leq 3000$.

Physical coherence of the simulated data was also verified through a local sensitivity analysis (gradient smoothness). The sensitivity of the pressure drop in the full pathways to airflow superficial velocity was evaluated using the partial derivative $\partial(\Delta P)/\partial U$, as shown in Fig. 7. The rates of pressure drop change in relation to velocity are substantial and have expected linear correlation trends. Expected linearity is based on the assumption of a turbulent regime of the flow within the packings; that is $\Delta P \propto U^2$ according to Forchheimer correlations (Ergun, 1952). Sensitivity to velocity is substantially higher in a turbulent regime, as expected, due to larger shear forces. In the laminar regime, noticeable fluctuations in the pressure gradient occur at high particle diameters. However, it should be noted that the relative variances are of similar magnitude, and the seemingly higher fluctuations are due to the logarithmic scale of the plot.

4.3. Data visualization

The pathways have variable cross-sections in the streamwise direction. Cross-sections vary smoothly with a periodicity equal to the particle diameter D_p. This variation for each pathway type can be represented with a corresponding area A_i ranging from its minimum value at particle centers to the maximum value in between. In simple cubic configurations of packed beds, these areas can be expressed via the particle diameter. For a full pathway, this area will range between $A_{1\ min} = D_p^2(1 - \pi/4)$ and $A_{1\ max} = D_p^2$. Half and quarter pathways will range as $A_{1/2} = \{A_{1\ min}/2, A_{1\ max}/2\}$ and $A_{1/4} = \{A_{1\ min}/4, A_{1\ max}/4\}$ correspondingly. This results in a 78.5% contraction of the pathways when passing through each particle layer.

In Fig. 8, one can see the velocity and pressure fields in the mentioned cross-sections. The sections were obtained near the penultimate particle layer to avoid vortex effects downstream of the final particle layer. Contour plots for all three pathway types are combined in Fig. 8 to facilitate comparison. Velocity fields showed an anticipated distribution both in minimal and maximum cross-sections. Pressure

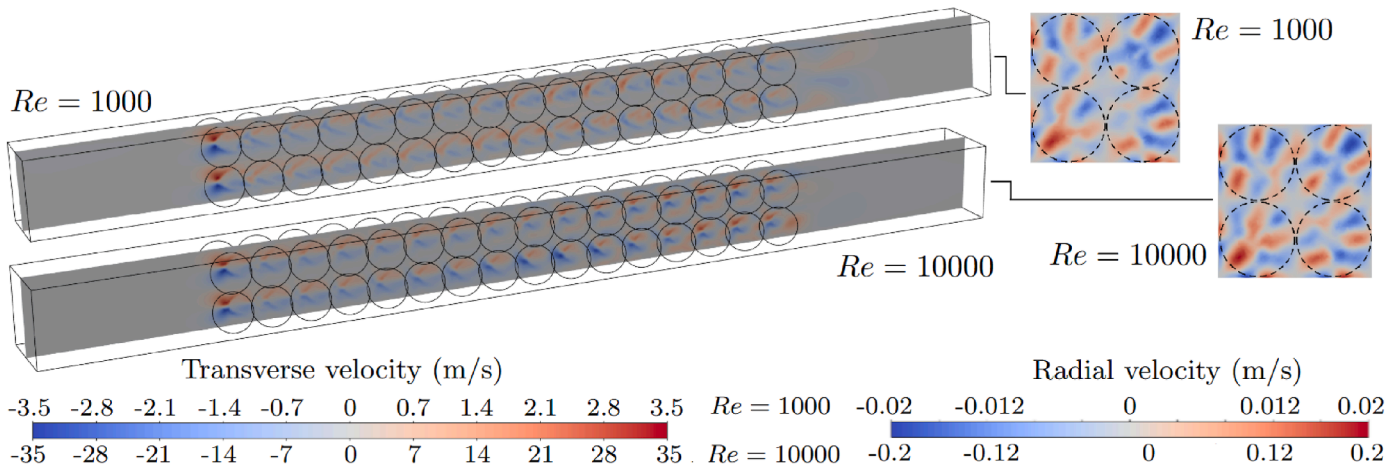


Fig. 5. Transverse and radial velocity profiles for a single design of the packing at Re = 1000 and Re = 10000: positive velocity values correspond to upward and outward flows.

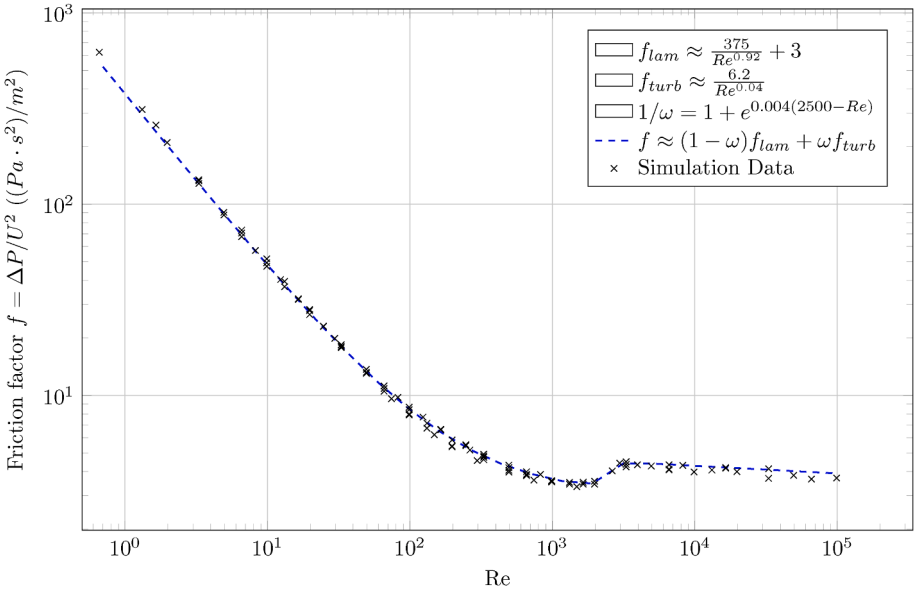


Fig. 6. Friction factor for a full pathway between the particles with diameters ranging from 1 to 15 mm.

fields were generally uniform, with star-shaped local rises behind the particles.

4.4. Data analysis and validation

The friction factor in laminar and turbulent regions can be estimated using simple correlations that might be merged into a single transitional form using sigmoid ω(Re). Friction factors for full, half, and quarter pathways can be represented with similar correlations:

$$f \approx (1-\omega) f_{lam} + \omega f_{turb}; \tag{9}$$

$$f_{lam} \approx \frac{k_1}{Re^{k_2}} + k_3; \tag{10}$$

$$f_{turb} \approx \frac{k_4}{Re^{k_5}}; \tag{11}$$

$$1/\omega = 1 + e^{0.004(2500-\mathrm{Re})}. \tag{12}$$

With an immutable sigmoid, laminar and turbulent components of the transition correlations are fitted with empirical coefficients. For example, for a full pathway, the following coefficients should be used: $k_1 = 375$, $k_2 = 0.92$, $k_3 = 3$, $k_4 = 6.2$, and $k_5 = 0.04$.

However, the pressure drop can be estimated using a normalized form $\Delta P' = \Delta P \cdot A/N_L$ where A is a coefficient that adjusts the pressure drop to a flow throughput G^v and N_L is the number of layers in a column. According to the continuity conservation law, this coefficient should represent a pathway cross-sectional area, i.e., $A \propto G^v/U$. For normalization purposes, any pathway cross-section (reentrant quadrilateral) can be reduced to an inscribed circle, as shown in Fig. 2. The circles inscribed within the reentrant voids of a dense orthogonal lattice of identical circles (particles) are proportional in size to the lattice circles. Thus, the pathway cross-section area can ultimately be reduced to the particle diameter:

$$\begin{cases} A_1 = D_p^2; \\ A_{1/2} = 0.5 D_p^2; \\ A_{1/4} = 0.25 D_p^2. \end{cases} \tag{13}$$

Correlations between the normalized pressure drop and Reynolds number for all considered types of pathways share the same trend.

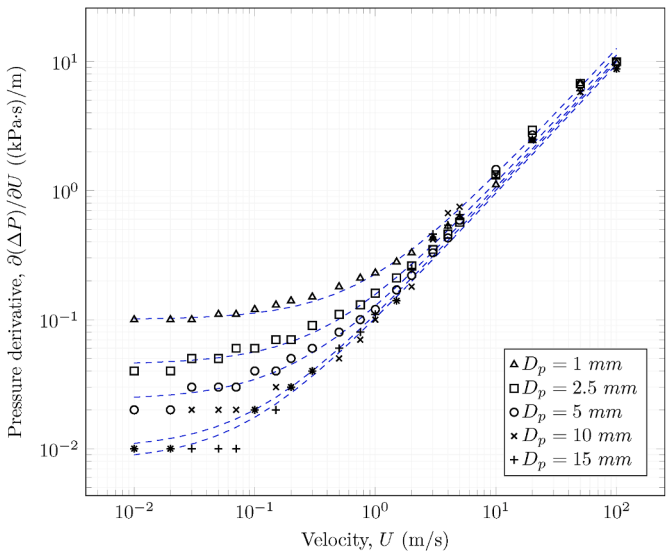


Fig. 7. Sensitivity analysis of pressure drop to flow superficial velocity.

Similarly to the friction factor, they were approximated with a transitional function, as shown in Fig. 9:

$$\Delta P' \approx (1-\omega)\Delta P'_{lam} + \omega \Delta P'_{turb}; \tag{14}$$

$$\Delta P'_{1\ lam} \approx a\mathrm{Re} + b\mathrm{Re}^2; \tag{15}$$

$$\Delta P'_{1\ turb} \approx c\mathrm{Re}^2; \tag{16}$$

$$1/\omega = 1 + e^{0.001(4500-\mathrm{Re})}. \tag{17}$$

Correlation coefficients for the transient approximation are listed in Table 2. Eq. (15) approximates the normalized pressure drop with acceptable accuracy, yielding an average absolute error of 7.2% with extremes of up to 20%. Correlation coefficients exhibit plausible trends due to variations in cross-sectional areas. However, these coefficients were obtained using a regression algorithm and were not derived from geometric analogy; thus, they are not suitable for trend derivation.

Eq. (9) captures the simulation data with reasonable accuracy, yielding a mean relative error of 3.34% and a maximum absolute error below 11%. Notably, the dispersion of simulation data is low, which ensures the validity of the proposed correlation for a wide range of particle diameters. Coefficient of variation *CV* was used to quantify the consistency of the data:

$$CV = \frac{\sigma}{\mu}100\%, \tag{18}$$

where $\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^2}$ is a standard deviation and $\mu = \frac{1}{N}\sum_{i=1}^{N}x_i$ is an arithmetic mean of values. This indicates a strong physical coherence of the data. The average coefficient of variation is 1.23% (with a maximum variation of 6.7%), indicating negligible dependence on particle diameter.

Summing the normalized pressure drops from each pathway, one can expand Eq. (3) into the direct correlation for engineering purposes:

$$\Delta P = \frac{N_L\left[(m-1)(n-1)\Delta P'_1 + 4(m+n-2)\Delta P'_{1/2} + 16\Delta P'_{1/4}\right]}{D_p^2(mn+m+n+1)}, \tag{19}$$

where $\Delta P'_i = f(\mathrm{Re})$ and $\mathrm{Re} = f(D_p)$. Thus, one can estimate the total pressure drop of a structured packed bed of spherical particles using this direct correlation, which requires the flow superficial velocity, basic fluid properties, and particle size and count.

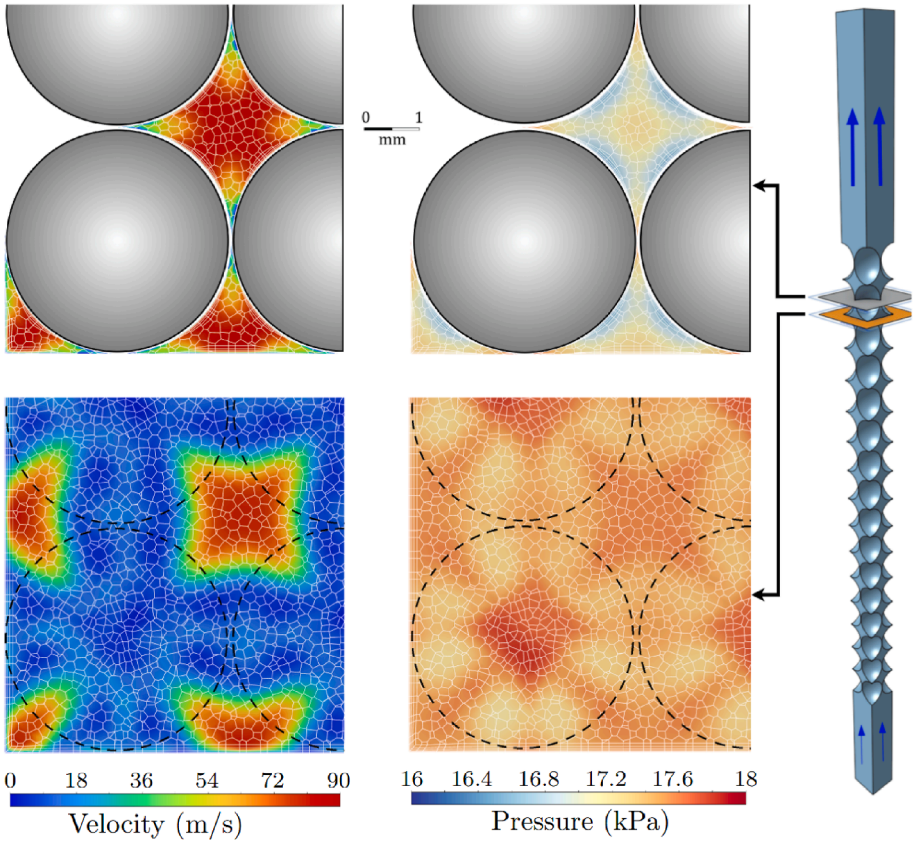


Fig. 8. Velocity and pressure fields in minimal (top row) and maximum (bottom row) cross-sections of the packing at $U = 15$ m/s.

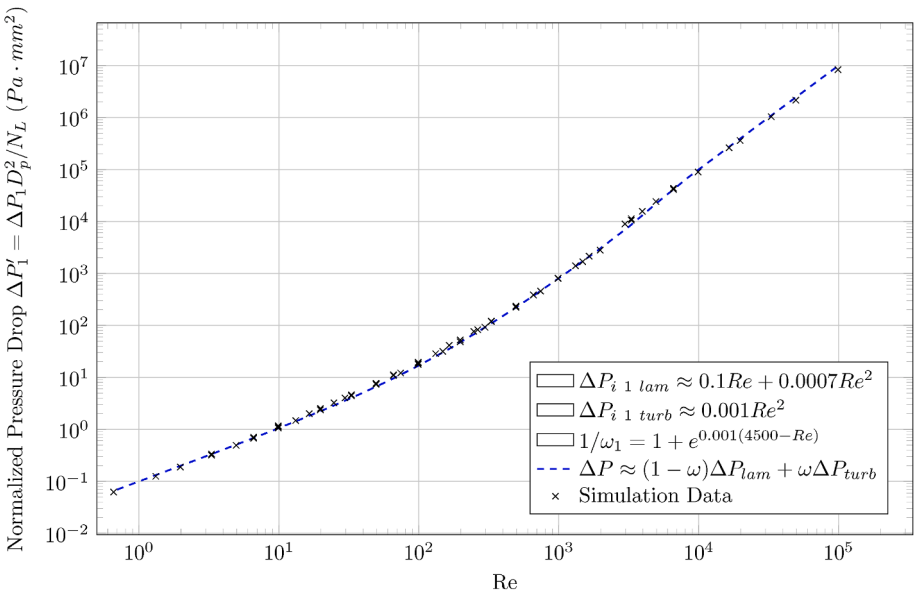


Fig. 9. Normalized pressure drop for a full pathway between the particles with diameters ranging from 1 to 15 mm.

Table 2
Correlation coefficients for normalized pressure drop function.

Pathway Type	*a*	*b*	*c*
Full	0.1	0.0007	0.001
Half	0.13	0.00055	0.0006
Quarter	0.13	0.0004	0.0004

Predictions of the obtained pressure drop correlation are in good agreement with the adopted experimental data (Bai et al., 2009; Calis et al., 2001; Tian et al., 2018; Wang et al., 2018; Yang et al., 2012). In Fig. 10, the normalized pressure drop was divided by the number of particles in each layer to visually separate the trends for different column sizes. Validation proved the consistency of the analytical solution across a wide range of Reynolds numbers that effectively account for major fluid properties (density and viscosity). Notably, the Reynolds number here doesn't include porosity and is based on particle diameter, as in Eq. (7). This makes calculations more straightforward and suitable for engineering purposes.

The details of validation are presented in Table 3. Here, the results were grouped by particle count in one layer of the packed bed column N_i. Superficial velocity U was derived from the available Reynolds values regarding fluid properties and reported porosity. For a strict arrangement of the spherical particles into a simple cubic configuration, the expected porosity is $\epsilon = 0.477$. However, for some considered data, different values were reported (e.g., $\epsilon = 0.3913$) due to certain specificities of the experimental setups. In such cases, experimental data were adopted with a corresponding correction. Discrepancies between analytical and experimental data were assessed using the modified mean absolute percentage error (MMAPE) to align with intuitive residuals:

$$\mathrm{MMAPE} = \frac{100\%}{n} \sum_{i=1}^{N} \frac{|\widehat{y_i} - y_i|}{y_i}, \tag{20}$$

where y are the measured (experimental) values and $\widehat{y}$ are those predicted by Eq. (19).

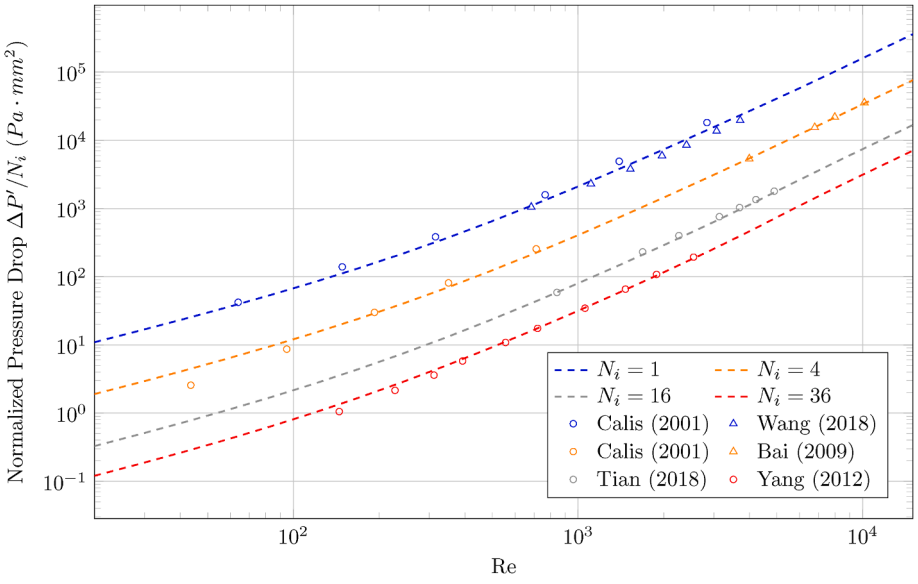


Fig. 10. Validation of the proposed correlation against adopted experimental data.

Table 3
Validation of the proposed correlation against adopted experimental data.

Case	Particle Count, N_i	Particle Diameter, D_p [mm]	Number of Layers, N_L	Superficial Velocity, U [m/s]	Medium	MMAPE [%]
(Calis et al., 2001)	1	12.7	8	0.05-2.08	Air	17.4
(Wang et al., 2018)	1	12	16	0.87-4.69	Air	15.9
(Calis et al., 2001)	4	12.7	16	0.03-0.52	Air	26.6
(Bai et al., 2009)	4	10	16	6.05-15.37	Air	1.4
(Tian et al., 2018)	16	13.55	40	0.05-0.29	Water	6.3
(Yang et al., 2012)	36	12	11	0.13-2.2	Air	8.7

5. Conclusion

This study demonstrates that the pathway analogy offers a robust and efficient framework for analyzing fluid flow and predicting pressure drops in structured packed beds with a simple cubic arrangement. The isolation and characterization of individual pathway types provided critical insight into the viability and constraints of applying lateral symmetry to reduce computational domain size. A key finding is the establishment of a universal correlation between the friction factor and Reynolds number, which holds independently of particle diameter and is effectively captured by a single transition function with an accuracy of 7.2% in a wide range of Reynolds numbers that correspond to practical cases: $1 \leq \mathrm{Re} \leq 10^5$.

The proposed pressure drop estimation method enables practical engineering applications by synthesizing these findings. The procedure involves first obtaining the pressure drop for each fundamental pathway type using the provided approximations based on the flow Reynolds number and column length. The total pressure drop is then calculated by summing the contributions of all pathways, weighted by their respective counts derived from the particle arrangement within a layer. This mechanistic, bottom-up approach facilitates fair predictions for beds of arbitrary sizes without numerical simulations. High-accuracy predictions require three numerical simulation runs (one for each type of pathway), which is still $mn+m+n-2$ times less compared to a full-scale numerical study for a column of $m \times n$ size. This provides a valuable tool for the design and scaling of structured packed bed systems.

Finally, while this paper establishes the pathway analogy for structured packings, the core concept could potentially be extended to other types of structured packed beds following additional parametric studies to classify their interstitial geometries.

CRediT authorship contribution statement

Maxim Nikitin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dmitry Pashchenko:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Nomenclature

Abbreviations

MMAPE	Modified Mean Absolute Percentage Error

Greek Symbols

ϵ	Porosity
μ	Arithmetic mean of the values [*]
μ_f	Fluid viscosity [Pa·s]
ν	Kinematic viscosity [m^2/s]
ρ_f	Fluid density [kg/m^3]
σ	Standard deviation of the values
ω	Specific turbulent dissipation rate [1/s]

Roman Symbols

ΔP	Pressure drop [Pa]
L	Column length [m]
G^m_{cross}	Cross flow rate between adjacent pathways [kg/s]
C_d	Discharge coefficient
N_i	Number of particles in each layer
k	Turbulent kinetic energy [m^2/s^2]
R_{ij}	Reynolds stress tensor [m^2/s^2]
I	Turbulent intensity [%]
Re	Reynolds number [−]
p	Pressure [Pa]
D_p	Particle diameter [m]
y^+	Dimensionless wall distance
Δx	Cell size [mm]
N_{cell}	Mesh cell count
U	Superficial flow velocity [m/s]

f	Friction factor [Pa·s^2)/m^2]
G^v	Volumetric flow rate [m^3/s]
N_L	Number of particle layers in a column

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